

Preview: planner_ablation_multifig.tex in an IEEETran two-column harness (labels sec:method and alg:skeleton are stubs)

I. METHODS (STUB)

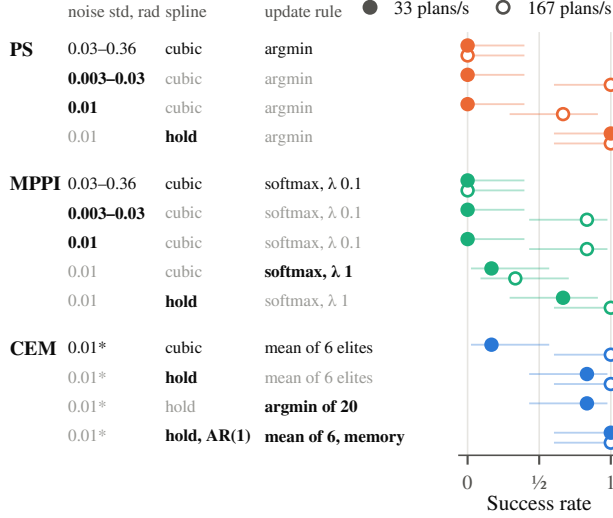
A. Planner Ablation

At the robot's joint gains and its 33 plans/s, CEM as shipped in MJPC completes the ladder (stand, lean, reach, release, four stand-back rungs) on 5/6 seeds and iCEM on 6/6; predictive sampling (PS) and MPPI complete 0/6, PS falling in the initial stand on five seeds and MPPI before or during the lean on every seed (Fig. 1a). Two of the three components that differ between the planners account for the gap. PS and MPPI scale their noise by the control range, $0.12 \times \frac{1}{2} \text{ctrlrange} = 0.03\text{--}0.36\text{rad}$ per joint (median 0.21), where CEM samples in radians with a 0.01rad floor; set to 0.01rad on the cubic spline MJPC's sampling planner ships with, PS and MPPI still complete 0/6 and 1/6. With the zero-order hold that CEM hard-codes, PS at 0.01rad completes 6/6 and MPPI 4/6, and CEM on the cubic spline drops to 1/6. Across twenty spline, knot-count and noise-colour configurations the separating quantity is how long the sampled perturbation of the executed action persists in the rollout (Fig. 1b): every configuration whose autocorrelation falls below 0.8 within 0.20s completes 0–2/6 and every one that holds it to 0.22s completes 5–6/6, for all four planners. A three-knot hold gives 0.33s by construction; iCEM's AR(1) knot noise gives it by correlation, which is why iCEM completes with either spline. The update rule (FOLD in Alg. 1) is not one of the two: with the hold and 0.01rad the argmin (PS, and CEM with one elite), the mean of $k \leq 10$ elites and the softmax mean all complete 4–6/6, and only $k = 20$, no selection at all, completes 0/6.

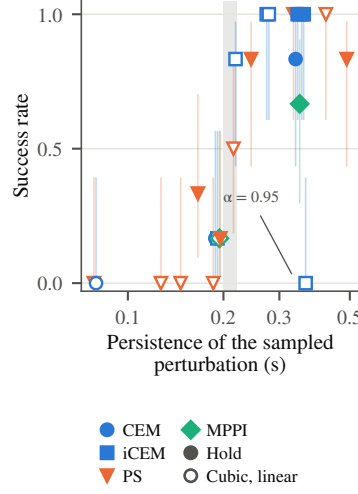
The update rule decides sensitivity. Over $\sigma \times \text{plan rate}$ (Fig. 1d) CEM completes $\geq 4/6$ in 16 of 20 cells, PS on the hold in 12, MPPI in 10 and PS on its shipped cubic spline in 5 of 15; at 33 plans/s CEM's admissible σ spans the tested decade and the other rules' one octave. The elite mean's random step is $\sigma/\sqrt{k}$ where the argmin's is a full σ , and that is what lets it trade noise for rate: at 0.02rad CEM completes 5/6 at 16.7 plans/s, where PS and MPPI gain nothing from the same change. The same asymmetry holds against model error (Fig. 1e; the plant's mass, joint stiffness or sliding friction is scaled after the planner has taken its model, 12 seeds per cell). Pooled over the three axes CEM completes 80/108, the argmin 47/108 and the softmax mean 42/108 (one-sided Fisher $p = 4 \times 10^{-6}$ and 1×10^{-7}). A plant two or three times as stiff as the model costs CEM nothing (11, 10/12) and PS 7–11 seeds, every loss a fall at

the lean onset, where the argmin executes one sample's full step; friction at $0.4\times$ costs CEM and iCEM nothing and PS and MPPI 5–6 seeds. Mass is the hard axis for every rule and the one on which iCEM fails (3/12 at $1.10\times$, every loss a fall in the stand: the heavier plant sags under the joint PD and iCEM's 3mrad step cannot walk the setpoints out of the sag); widening its floor to 0.03rad restores 9/12 (Fig. 1f), and the same widening costs CEM nothing at baseline (11/12) and PS three seeds. This is tolerance to parametric plant error; estimator noise and unmodelled actuator dynamics are not in the bench.

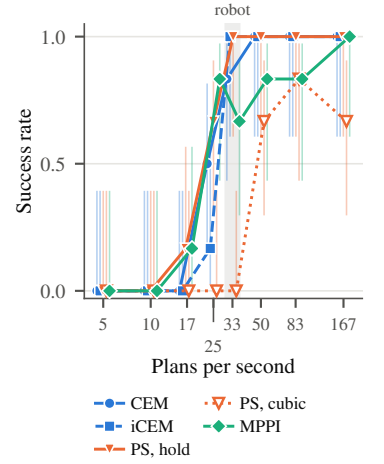
The rate floor sits just below the robot's rate (Fig. 1c): at 0.01rad, 25 plans/s completes 1–5/6, 16.7 plans/s 0–1/6 and 10 plans/s 0/6, every failure a backward fall 2–3s into the lean, while the 12s stand survives at 5 plans/s; this is the 22Hz desktop failure of Sec. I. The executed target step per plan depends on σ and the rule (CEM 5–8mrad at every rate from 167 down to 5 plans/s), so setpoint speed is $\text{step} \times \text{rate}$, and the lean onset needs 30–40mrad/s of directed setpoint motion (RMS over the 27 joints), which 33 plans/s just supplies. Whole-body locomotion MPC runs at 50–100 plans/s because it re-decides a contact set every ≈ 0.3 s of stance and stabilises an underactuated phase with the pendulum's time constant; supported reaching has a fixed contact set, a posture the 500Hz joint PD holds between plans, and one fast event. The planner is a setpoint optimiser over a stabilised plant, its rate requirement is set by how fast the setpoints must move, and a larger sampling step buys rate back only for the elite mean. Planning from a state 100ms stale costs nothing at 33 plans/s and 200ms costs 2–5 seeds, or none with the deploy node's forward prediction; halving the horizon to 0.5s completes 0/6 and quartering the rollouts 0–2/6, so the deployed 1s, 20-rollout configuration is close to minimal in every direction but the noise.



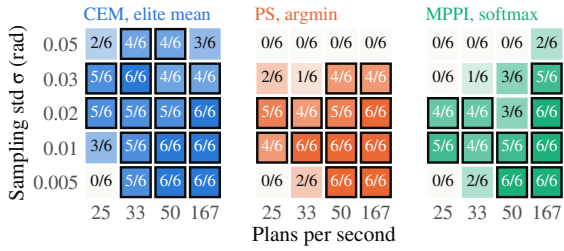
(a) From PS to CEM one component at a time; the cell that changed from the row above is set in black. Filled: 33 plans/s, hollow: 167.



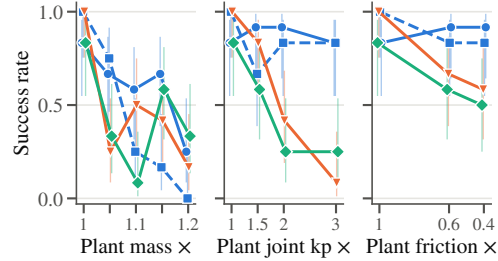
(b) Success at 33 plans/s against how long the sampled perturbation of the executed action persists in the rollout (lag at which its autocorrelation falls below 0.8), for 20 spline, knot-count and noise-colour configurations.



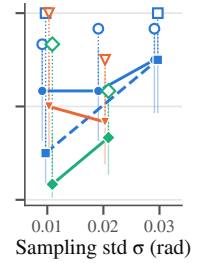
(c) Success against plan rate at $\sigma = 0.01$ rad.



(d) Success over σ and plan rate for each update rule on the hold spline; outlined cells are at $\geq 4/6$.



(e) Success at 33 plans/s when the plant's mass, joint stiffness or sliding friction is scaled and the planner's model is not; 12 seeds, lines as in (c).



(f) Wider σ at baseline (hollow) and under $1.10\times$ mass (filled); 12 seeds.

Fig. 1: Sampling-based planner ablation: components, sensitivity, and model error. Simulation on the robot's joint gains; six seeds per point unless stated, Wilson 95% intervals; hold spline, $\sigma = 0.01$ rad, 20 rollouts, six elites and $\lambda = 1$ unless varied. 0.01*: CEM and iCEM floor the elite standard deviation at 0.01 rad and refit at $1.3\times$ the floor.